DSA Practical No: 6

Group C- Assignments (Any TWO) 6 In any language program mostly syntax error occurs due to unbalancing delimiter such as (), {}, []. Write a program using stack to check whether a given expression is well parenthesized or not.

Code

```
#include <iostream>
using namespace std;
#define size 10

class stackexp {
    int top;
    char stk[size];
public:
    stackexp() {
        top = -1;
    }
    void push(char);
    char pop();
    int isfull();
    int isempty();
};

void stackexp::push(char x) {
    top = top + 1;
    stk[top] = x;
}

char stackexp::pop() {
    char s;
    s = stk[top];
    top = top - 1;
    return s;
}

int stackexp::isfull() {
    if (top == size)
        return 1;
```

```
    else
        return 0;
}

int stackexp::isempty() {
    if (top == -1)
        return 1;
    else
        return 0;
}

int main() {
    stackexp s1;
    char exp[20], ch;
    int i = 0;
    cout << "\n\t!! Parenthesis Checker..!!!!" << endl;
    cout << "\nEnter the expression to check whether it is well-formed or not: ";
    cin >> exp;
    if ((exp[0] == ')' || exp[0] == ']' || exp[0] == '}')) {
        cout << "\nInvalid Expression.....\n";
        return 0;
    } else {
        while (exp[i] != '\0') {
            ch = exp[i];
            switch (ch) {
                case '(': s1.push(ch); break;
                case '[': s1.push(ch); break;
                case '{': s1.push(ch); break;
                case ')': s1.pop(); break;
                case ']': s1.pop(); break;
                case '}': s1.pop(); break;
            }
            i = i + 1;
        }
    }
    if (s1.isempty()) {
        cout << "\nExpression is well parenthesized...\n";
    } else {
        cout << "\nSorry!!! Invalid Expression or not well parenthesized....\n";
    }
    return 0;
}
```